# AI-Driven Legal Judgement Prediction

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# Problem Statement

India's legal system struggles with efficiently analyzing case data, making decision-making slow and complex for both professionals and the public. There is a need for an intelligent solution that can process legal information accurately and provide valuable insights.

# Literature Survey

| S.No | Title | Dataset | Advantages | Disadvantages |
|---|---|---|---|---|
| 1 | A Systematic Literature Review of Machine Learning Methods in Predicting Court Decisions | 22 selected studies from Scopus and Web of Science databases. | Identifies trends in legal prediction and highlights key decision factors. | Limited scope and focus on binary classification |
| 2 | A Predicting Outcome of Judicial Cases and Analysis Using Machine Learning | Indian judicial case data collected from various legal websites. | Improves decision-making and enhances accessibility to legal insights. | potential bias and varying accuracy across different models. |

# Literature Survey (contd.)

| S.No | Title | Dataset | Advantages | Disadvantages |
|---|---|---|---|---|
| 3 | Using Machine Learning to Predict Decisions of the European Court of Human Rights | Judgments from the European Court cases collected from the HUDOC database. | Identifies key factors and influencing decisions. | Lower accuracy for future case predictions and limited generalization, |
| 4 | Legal Judgment Prediction System | 100 Indian murder case judgments collected from the Supreme Court of India. | Helps in predicting IPC sections and assists legal professionals | Limited dataset size and potential bias, |

# Research Gap

Most legal judgment prediction studies focus on Western legal systems, leaving Indian law underexplored. Existing models primarily predict binary outcomes but struggle with detailed case classifications, legal provisions, and contextual nuances. Many also lack explainability, making AI-driven insights hard to trust. Additionally, limited research integrates multi-court datasets from Supreme, High, and District Courts, which is crucial for improving accuracy and real-world applicability in India's legal system.

# Motivation

India has over 5 crore pending cases, leading to prolonged delays and an overburdened legal system. Many people struggle to navigate complex legal proceedings, while courts face challenges in managing vast amounts of case data. With advancements in AI and machine learning, there is an opportunity to analyze past judgments and predict case outcomes, helping streamline legal processes. This project is motivated by the need to reduce delays, assist legal professionals, and enhance public understanding of the judicial system, ultimately making justice more efficient and accessible.

# Abstract

India's legal system struggles with case backlogs and complex proceedings. This project develops an AI-powered tool to analyze past judgments and predict case outcomes, applicable laws, and penalties. Using machine learning and natural language processing, it processes legal texts to provide insights with promising accuracy. The goal is to assist legal professionals and improve public understanding, making the judicial process more transparent and efficient.

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